A RING MAP WHICH BECOMES AN ISOMORPHISM AFTER INVERTING P IS A UNIVERSAL HOMEOMORPHISM EXACTLY WHEN ITS KERNEL IS LOCALLY NILPOTENT AND EVERY ELEMENT IS BRIDGED BY A P-POWER MULTIPLE AND A P-POWER

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Following the multi-claim convention for this source, source_url is the source-level URL https://stacks.math.columbia.edu/ and the source label lemma-universal-homeo-iso-if-invert-p is carried in claim_key and here rather than in a per-claim tag URL.

Editorial note. The preamble is the chapter's shared Stacks Project preamble, retained as sources/stacks-morphisms/retained-preamble.tex: a line-for-line excerpt of the shared Stacks preamble staged with this snapshot. No macro was invented or redefined, and this card adds no preamble text of its own.

Editorial note. Statement span 11825-11837 and proof span 11839-11870 come from the halfA pool card operations/pool-cards/stacks-morphisms-halfA.json and were re-read in the pinned chapter file before this card was written.

Editorial note. The statement's parenthesis giving the example of a nilpotent p is the author's and is preserved.

Editorial note. The proof names the candidate subalgebra B' by an explicit set-builder expression; the display is preserved unchanged.

Editorial note. 4 cross-references occur inside the counted spans; they are listed in external_references as navigation pointers only.

Allowed standard prerequisites (editorial).

- The algebra criterion that a ring map whose kernel is locally nilpotent and which has the stated bridging property gives a universal homeomorphism. Cited at proof lines 11839-11840. Gives the easier implication from the algebraic condition to the geometric one.
- If a morphism of spectra is a universal homeomorphism then the kernel of the corresponding ring map consists of nilpotent elements, cited from the algebra chapter. Cited at proof lines 11841-11843. Allows the base ring to be replaced by the image, so that the map is injective.
- The set of elements bridged by a p-power multiple and a p-power is a subalgebra, cited from the algebra chapter. Cited at proof lines 11845-11849. Produces the subalgebra whose equality with the target is what has to be shown.

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- The two-step criterion producing an element outside a subalgebra with either two consecutive powers inside or a prime multiple and a prime power inside, cited from the same chapter. Cited at proof lines 11851-11854. Supplies the element used to derive the contradiction that closes the proof.

Source reference. Original lemma 0545: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6628–6628. Same-chapter reference used in the retained author text at source line(s) 11844. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 00FL: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 575–575. Same-chapter reference used in the retained author text at source line(s) 1301, 11842, 11946, 12203, 14779. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 00I5: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 667–667. Same-chapter reference used in the retained author text at source line(s) 11820, 11840, 11973. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 0CNC: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 15887–15887. Same-chapter reference used in the retained author text at source line(s) 11807, 11851, 12205. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Lemma 0.1. *Let p be a prime number. Let $A \to B$ be a ring map which induces an isomorphism $A[1/p] \to B[1/p]$ (for example if p is nilpotent in A). The following are equivalent*

(1) $\mathrm{Spec}(B) \to \mathrm{Spec}(A)$ *is a universal homeomorphism, and*
(2) *the kernel of $A \to B$ is a locally nilpotent ideal and for every $b \in B$ there exists a p-power q with qb and b^q in the image of $A \to B$.*

Proof. If (2) holds, then (1) holds by Algebra, Lemma 00I5. Assume (1). Then the kernel of $A \to B$ consists of nilpotent elements by Algebra, Lemma 00FL. Thus we may replace A by the image of $A \to B$ and assume that $A \subset B$. By Algebra, Lemma 0545 the set

$$B' = \{b \in B \mid p^n b, b^{p^n} \in A \text{ for some } n \geq 0\}$$

is an A-subalgebra of B (being closed under products is trivial). We have to show $B' = B$. If not, then according to Lemma 0CNC there exists a $b \in B$, $b \notin B'$ with either $b^2, b^3 \in B'$ or there exists a prime number ℓ with $\ell b, b^\ell \in B'$. We will show both cases lead to a contradiction, thereby proving the lemma.

Since $A[1/p] = B[1/p]$ we can choose a p-power q such that $qb \in A$.

If $b^2, b^3 \in B'$ then also $b^q \in B'$. By definition of B' we find that $(b^q)^{q'} \in A$ for some p-power q'. Then $qq'b, b^{qq'} \in A$ whence $b \in B'$ which is a contradiction.

Assume now there exists a prime number ℓ with $\ell b, b^\ell \in B'$. If $\ell \neq p$ then $\ell b \in B'$ and $qb \in A \subset B'$ imply $b \in B'$ a contradiction. Thus $\ell = p$ and $b^p \in B'$ and we get a contradiction exactly as before. $\square$